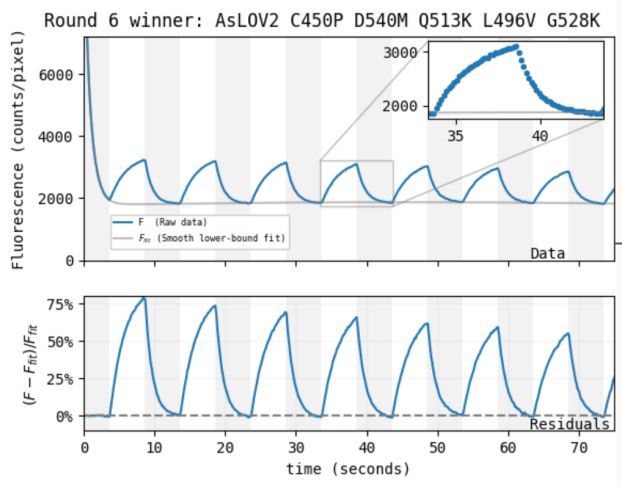
Expression, Purification and Crystallization of Magnetically Controllable Protein

As of 8/12/2024

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Wakatsuki Lab, Stanford Structural Biology - SSRL, SLAC

An Introduction to MagLOV



might mean magnetic control of all of these

LOV-system	Effectors	Fold activation	Purpose	Lifetime (s)	References
AsLOV2	**Phototropism**			**2–4300 s (*WT* = 55 – 81)**	**a**
LINuSs1	NLS	3–7	Nuclear localization[20]	240 s (4 min)	Yazawa et al., 2009
LOV-TAP	TrpR	6, 70	DNA-binding/Tryptophan repressor	NA	Strickland et al., 2008; Wu et al., 2011
LOV-Rac	Rac1 (GTPase)	10	Control actin cytoskeletal dynamics	43 s	Wu et al., 2009
Tulips	ePDZ	2–49	Peptide caging		Strickland et al., 2012
LOV-TetR	TetR	NA	Tetracycline/DNA-binding	30 s	Moon et al., 2014
YtvA	**Stress**			**72–16000 (*WT* = 6240)**	**a**
Dusk/Dawn	FixL/FixJ	460	Transcription	NA	Ohlendorf et al., 2012
YF1	FixL	68	Kinase activity	~5900	Moglich et al., 2009
TetR	TetR	NA	Tetracycline/DNA-binding	2700	(100)
EL222	**Transcription**			**2.7–2000 (*WT* = 29)**	**a**
EL222-TF	HTH	>108	Transcription	~30 s	Motta-Mena et al., 2014
VVD	**Circadian clock**			**18000 s (WT)**	**a**
Caspase-9	Homo-dimerization	7.6–21	Caspase9 activation to regulate apoptosis	NA	Nihongaki et al., 2014
GAVPO	Gal4	200–300	Light induced transactivation of Gal4	7200 s	Chen et al., 2013; Ma et al., 2013
Magnets	Selective Dimerization of VVD	40-fold estimate	Create VVD heterodimers of two components	25 s–17000 h	Kawano et al., 2015
FKF1	**Flowering**			**>100000 s (WT)**	**a**
LITEZ	2-hybrid	53	Transcriptional control	NA	Polstein and Gersbach, 2012
LAD	Light induced dimerization	5	Rac1 induction of lamellipodia	62 h	Yazawa et al., 2009

*a, refer to **Table 2** for more details on range of photocycle lifetimes.*

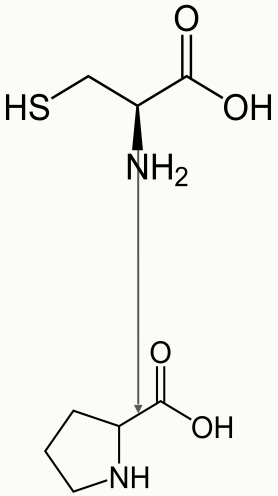
MagLOV has been confirmed to exhibit its effects in E.coli colonies and in purified protein

And that’s it unfortunately

MagLOV Mutations(against wild type AsLOV2)

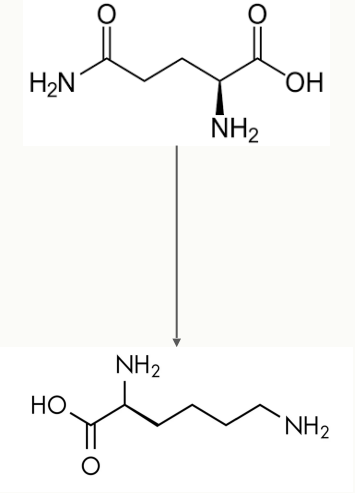
C450P L496V Q513K D540M G528K

C450 (AsLOV2)



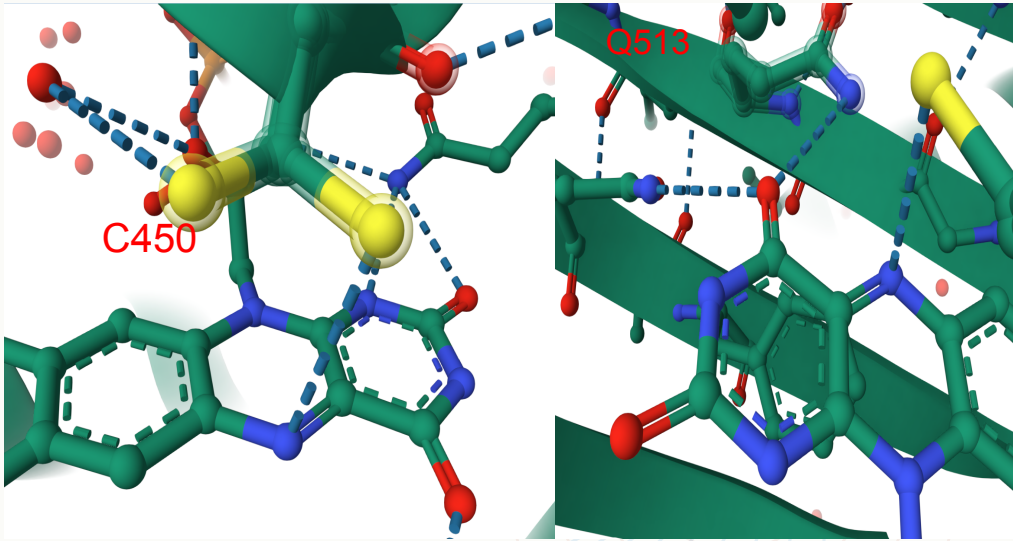
P450 (MagLOV)

Q513 (AsLOV2)

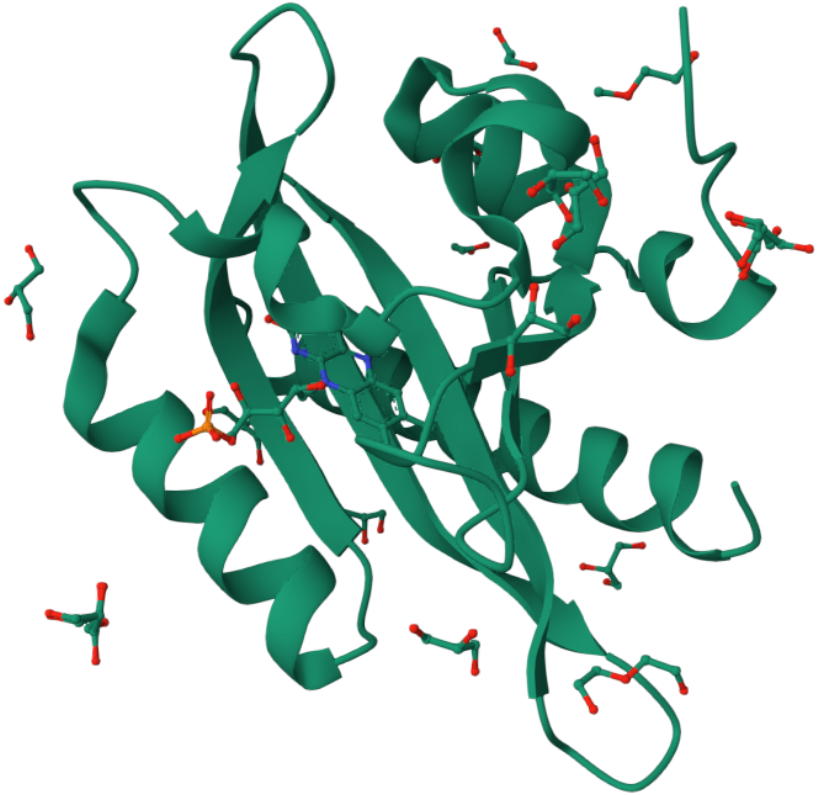


K513 (MagLOV)

Two critical residues are mutated (PDB 7PGX)

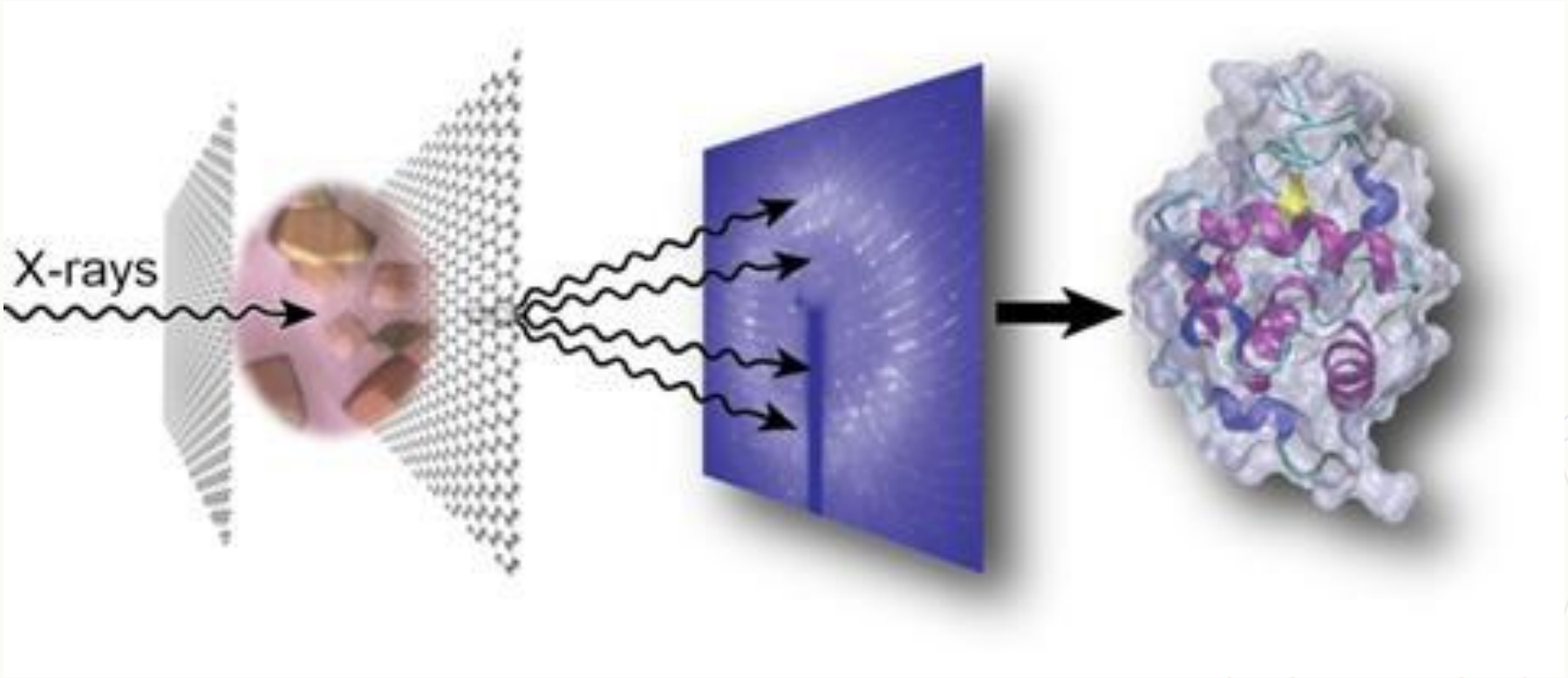


SLAC/Wakatsuki Lab



Research Question: How is it that MagLOV is able to fluoresce, and how is it’s fluorescence suppressed by a magnetic field?

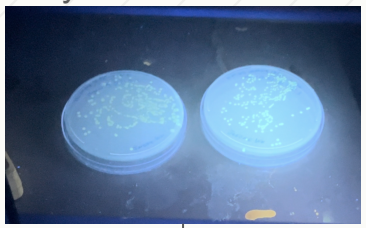
SLAC/Wakatsuki Lab



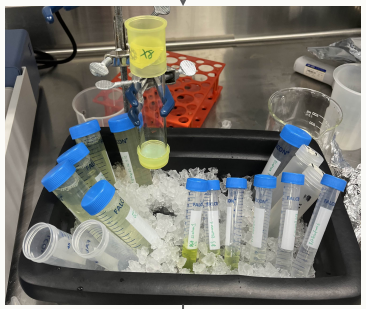
General Timeline

SLAC/Wakatsuki Lab

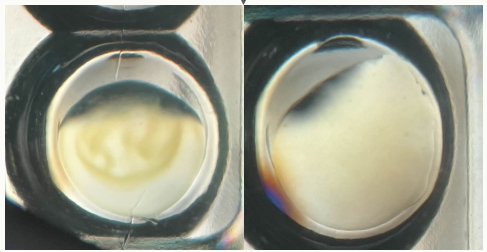
Crystal Production



Optimization of Protein Expression in E.Coli
Complete!



Protein Purification in Nickel NTA Resin (pictured) And FPLC (not pictured)
Complete!



Crystals
Pending…

MFE Measurements

Colony Check (reproducing previous work)
Complete!

Purified Protein Check (reproducing previous work)
Complete!

Crystal Check
Pending…

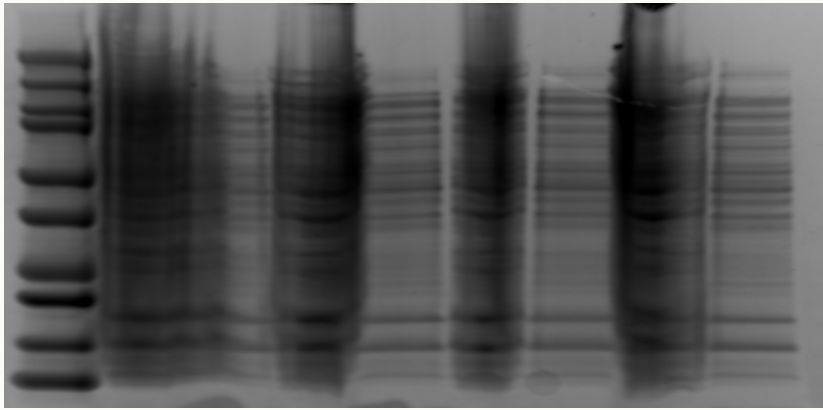
Protein Expression in E.Coli

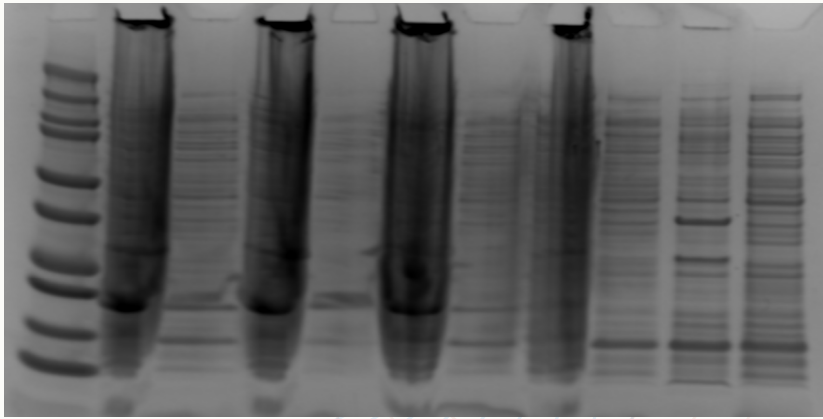
pRSET Plasmid—Leaky.

- Requires high selective pressure to ensure Plasmid stability
- Generally low yields (0.1mg/mL on low vol)
- Grows very slow

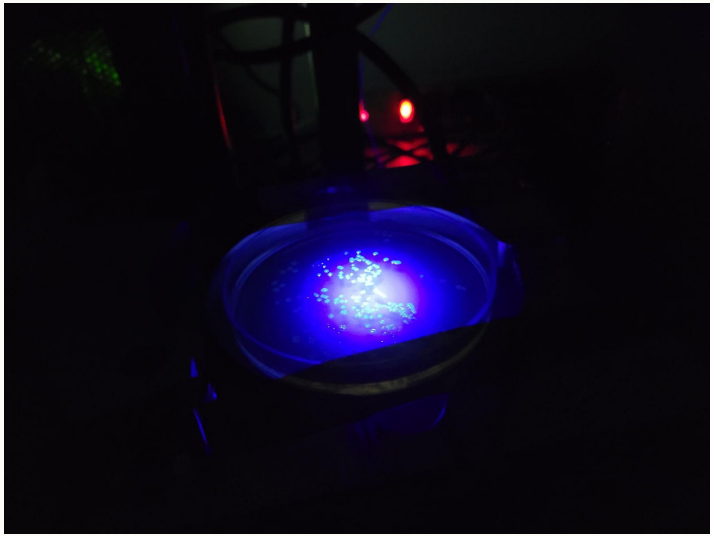
PET28a1+ Plasmid—Not Leaky

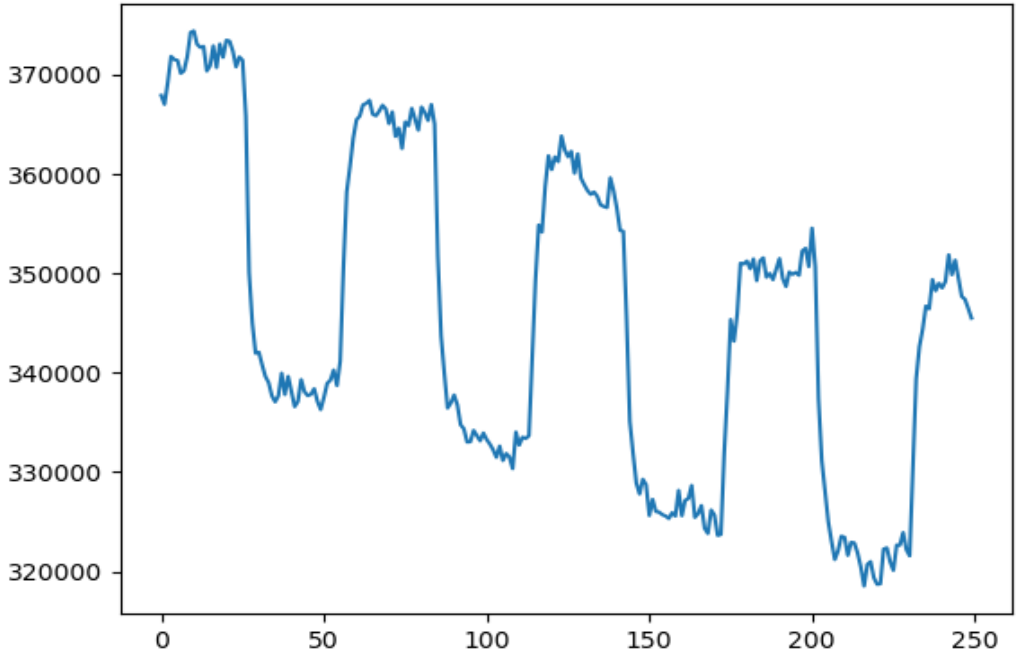
- Fast growth
- High yield (0.5mg/mL on low volume, 10-22 mg/mL on high vol)





MFE Measurements From Colonies (pRSET Plasmid)





Ni-NTA Purification

5mL Beads for 50 mL Lysate Supernatant.

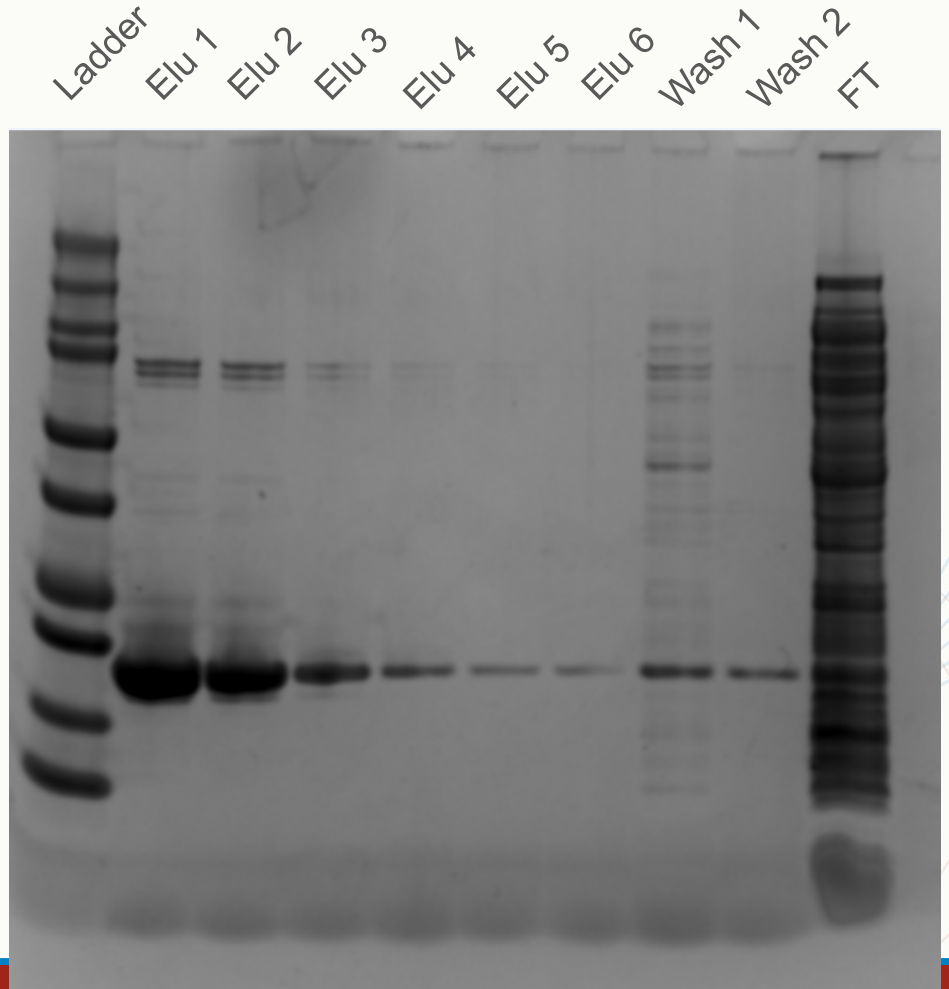
10 mL Lysis buffer to equilibrate beads

50 mL Lysate Supernatant Added. Rolled at 40 rpm for 30 minutes. Flowthrough Collected (W10)

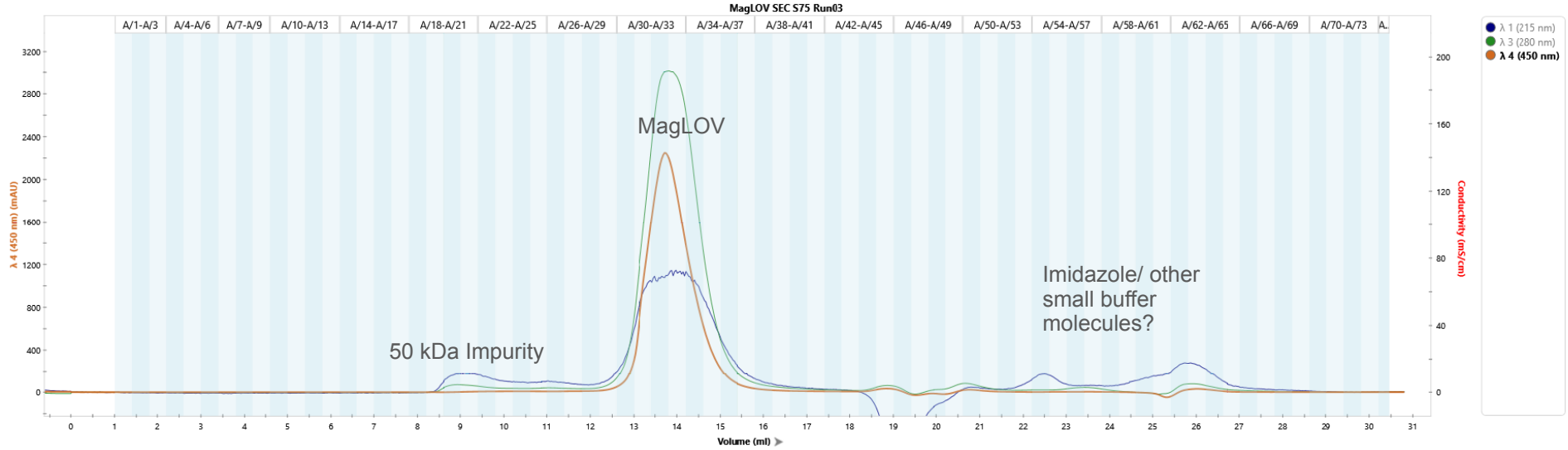
4 washes of 50 mL wash buffer. Washes 1 and 4 were run (W8, W9)

6 5 mL fractions of elution Buffer (W2-7)

Elutes 1-3 were concentrated into a single 1.5mL tube using a 10 kDa 15 mL Millipore Centrifuge Filter Tube



SEC



Wells of interest:
A20, A22 (impurity)
A30, A32, A33, A34, A35, A36 (MagLOV)

FPLC Gel

Loaded Volume of each was

W1 Ladder

W2 Ni-NTA Elute (concentrated)

W3 Blank to prevent potential bleed

W4 A20

W5 A22

W6 A30

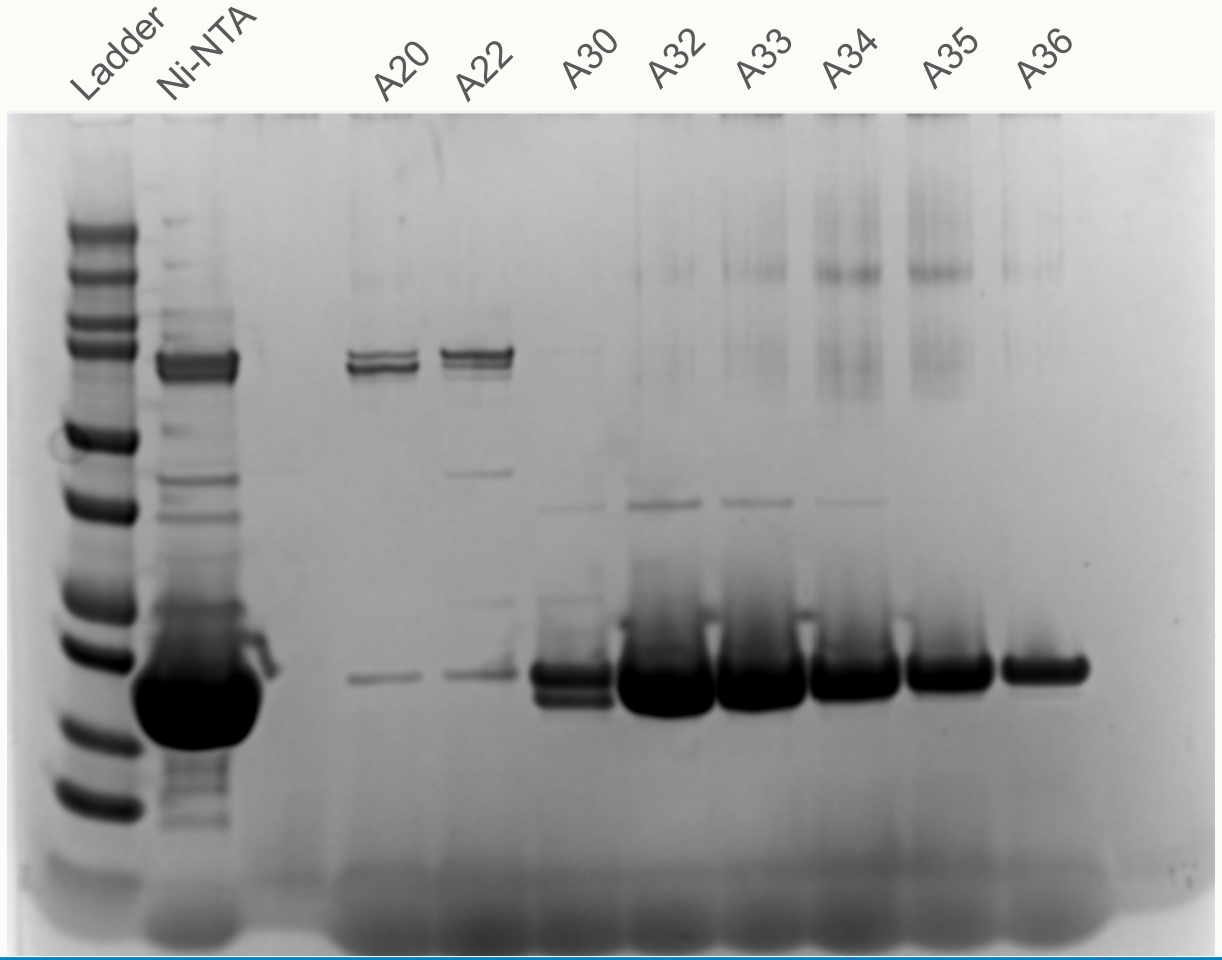
W7 A32

W8 A33

W9 A34

W10 A35

W11 A36



Current Work: Crystallization

Conditions:

1) 6% PEG4000, 4.5pH NaOAc, 30% Gly
2) 6% PEG4000, 4.8pH NaOAc, 30% Gly
3) 6% PEG4000, 5.0pH NaOAc, 30% Gly
4) 7% PEG4000, 4.5pH NaOAc, 30% Gly
5) 7% PEG4000, 4.8pH NaOAc, 30% Gly
6) 7% PEG4000, 5.0pH NaOAc, 30% Gly
7) 8% PEG4000, 4.5pH NaOAc, 30% Gly
8) 8% PEG4000, 4.8pH NaOAc, 30% Gly
9) 8% PEG4000, 5.0pH NaOAc, 30% Gly

Top Drop: 11mg/mL

Middle Drop: Empty

Bottom Drop: 22mg/mL

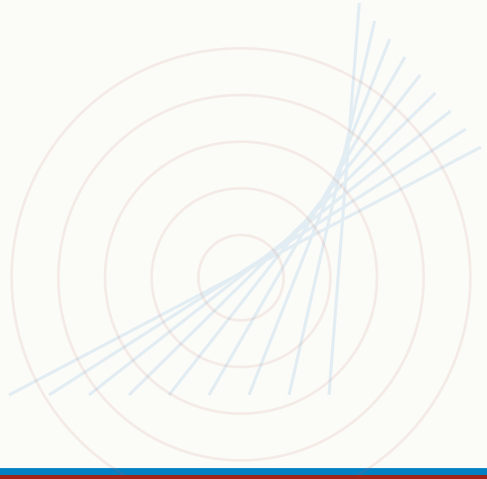
NO HITS (as of 8/12/24)

Future Work

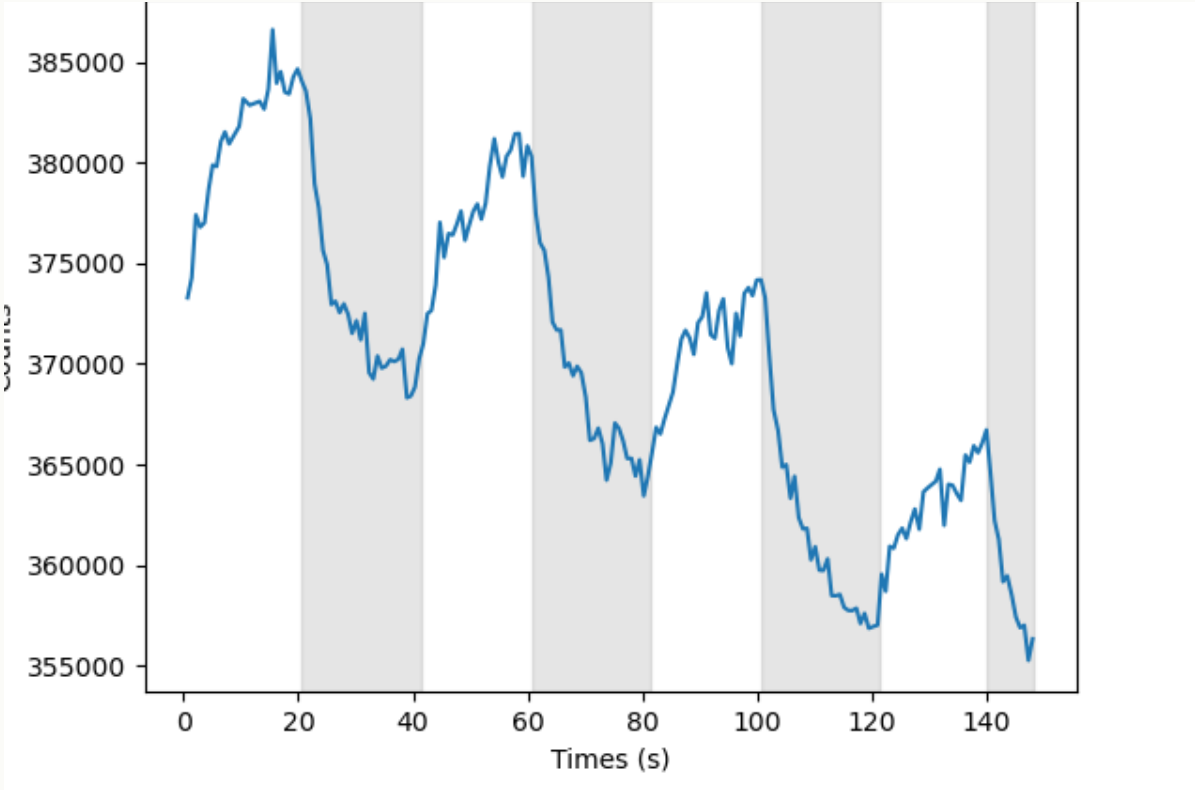
Diffraction!!!

Assays!!!!

NMR?



MFE Measurement From Purified Protein (pRSET Plasmid)



Acknowledgements

Wakatsuki Lab—Soichi Wakatsuki, Samsuzzoha Mondal, Dara Dowlatshahi

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Bio-X

Questions?